

15 Years of Progress

Where Should UNICEF Invest Next?

Life Expectancy Analysis across 193 Nations | 2000-2015

Setting the Stage

The Year is 2016

A newly appointed UNICEF Director is reviewing 15 years of global health data (2000 - 2015) to decide where to allocate aid.

Our analysis: Which countries made the biggest gains in life expectancy, what drove those gains, and where are the gaps that still demand attention?



The Problem

Progress was uneven. Some nations surged ahead while others are still behind



Why 15 years?

2008 financial crisis followed by the Euro crisis in 2012 has set back many countries. Analysis on a longer time horizon is needed



Dataset

WHO data across 193 countries, enriched with HDI and PM2.5 data from UNDP & World Bank



The Big Picture

How much did life expectancy improve across nations from 2000 to 2015?

Life Expectancy: A Global Snapshot

66.8 → 71.6

2000

2015

Global avg
life expectancy

+5.2 yrs

Average gain in
developing nations

51

Sierra Leone

88

Slovenia

Life Expectancy,
Highest & Lowest - 2015

Key Observations



Biggest movers: Eritrea (+19 yrs), Rwanda (+18 yrs), Botswana (+18 yrs)



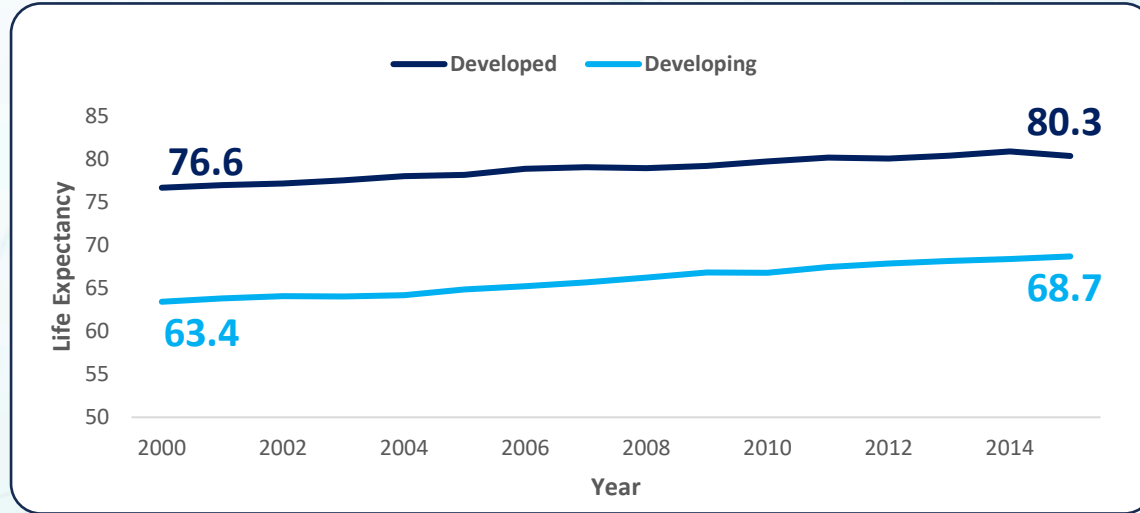
49 countries had life expectancy below 60 in 2000 - by 2015, that number dropped to 21



Countries in Sub-Saharan Africa still trail the global average by over a decade

The Development Divide

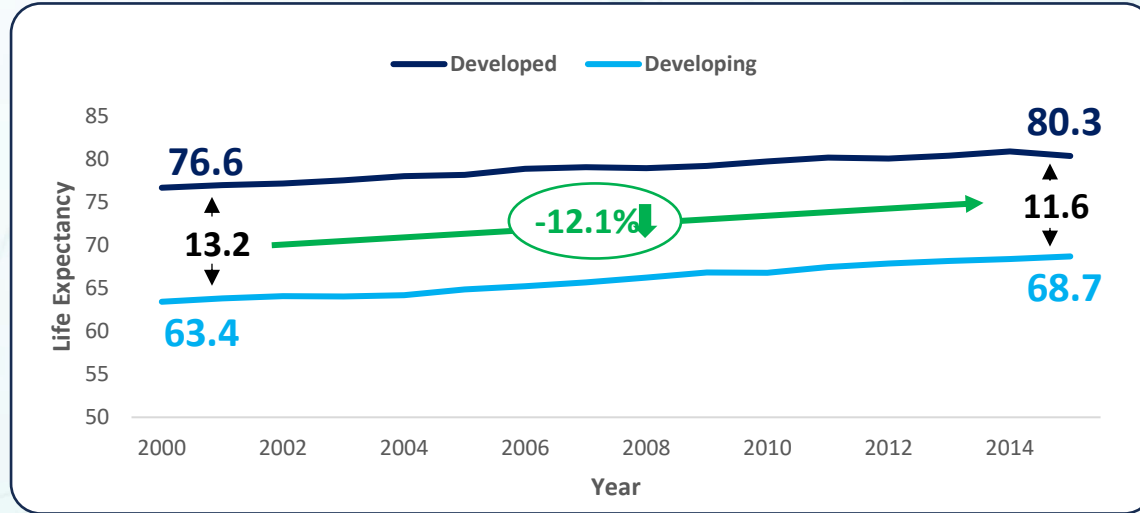
Average Life Expectancy over the years



- Gap between developing vs developed nations shows -> health “**systems**” problem still exists
- Progress is happening, but at this rate, closing the **~11-year gap would take over a century**

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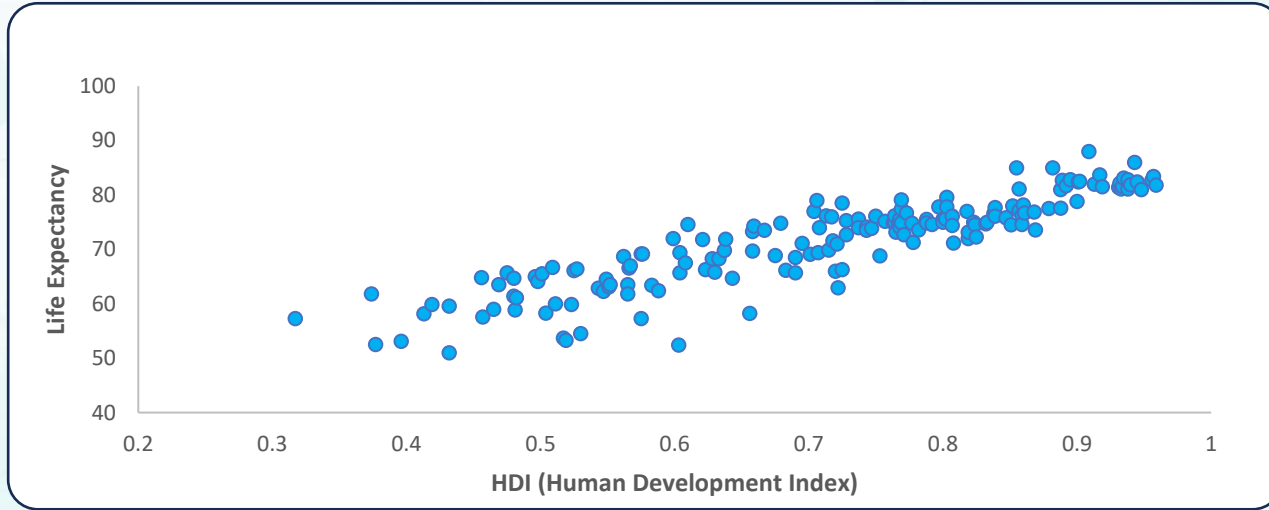
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Money Doesn't Buy Health

Life Expectancy vs HDI

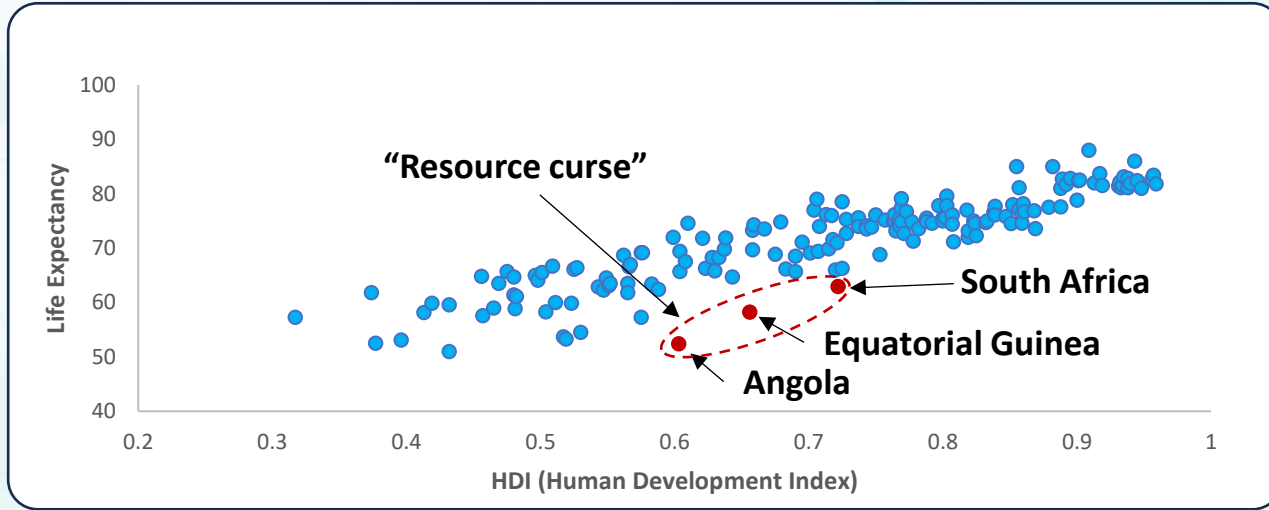


$$\begin{array}{c} \text{HDI} \\ = \\ \text{Life Expectancy} \\ + \\ \text{Schooling} \\ + \\ \text{GNI} \end{array}$$

- Angola, Equatorial Guinea have higher HDI (Oil production income) but lower health
- South Africa – one of the strongest economies in Africa has high HIV/AIDS deaths

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What Drove the Gains

Which factors correlate most with life expectancy improvements?

Factors Affecting Life Expectancy

Correlation Analysis (From Highest to Lowest)

Schooling	Income Composition of Resources	BMI	Thinness	GDP/capita	Alcohol	PM2.5	Percentage Expenditure	Total Expenditure
0.75	0.72	0.57	-0.47	0.46	0.4	-0.39	0.39	0.22



Schooling

0.40 0.69
Developed Developing

Strongest positive correlation. Each additional year of schooling -> **~2-year increase** in life expectancy.



Income Composition

0.72 0.64
Developed Developing

Income composition shows a **clear positive trend** - countries with **better resource** distribution live longer.

Factors Affecting Life Expectancy



BMI

-0.03

Developed

0.54

Developing

BMI in **developing nations** acts as a **food security** indicator. However, this does not capture the **individual-level risks of obesity** in **developed nations**



GDP/Capita

0.35

Developed

0.39

Developing

Nations like Uruguay, Antigua, Oman, Barbados have smaller GDP/Capita compared to EU & GCC countries but are offering **more targeted health systems**.



Pollution Levels

-0.56

Developed

-0.26

Developing

Countries in GCC and Asia show **higher levels of pollution** but still manage good life expectancy. Not every country is able to offset pollution damage



Expenditure

0.35

Developed

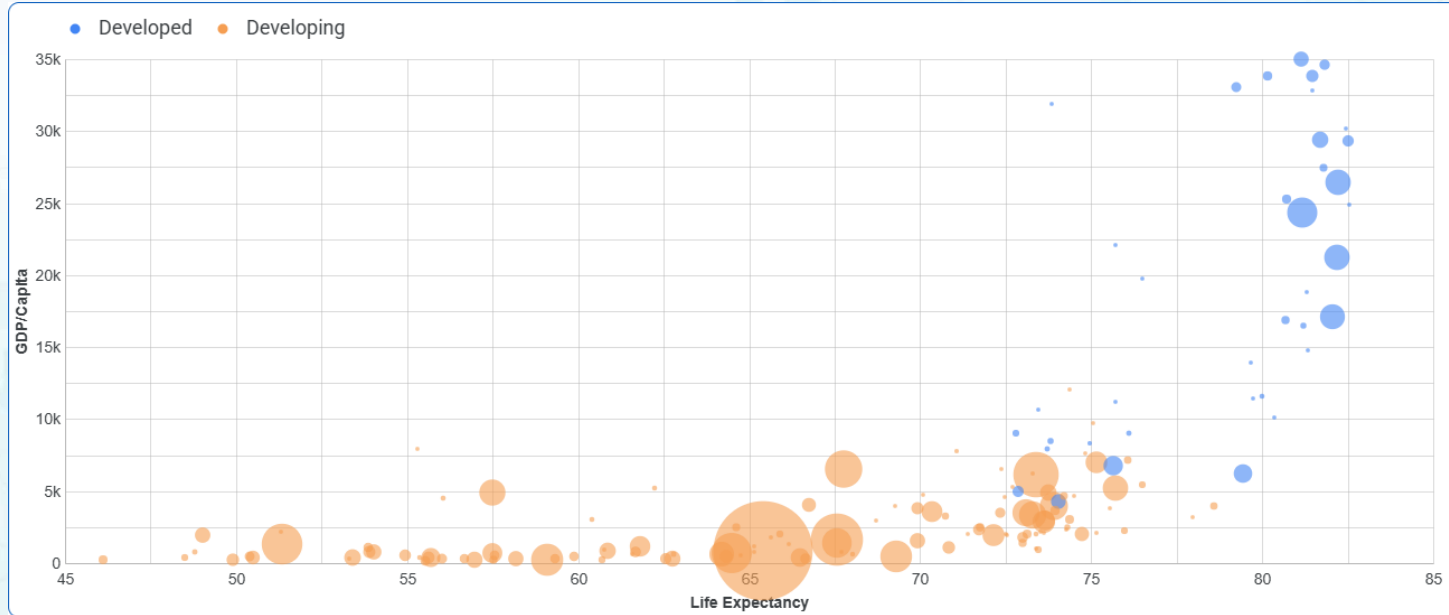
0.34

Developing

Correlation analysis shows that it's not about what share of budget goes to health, but **the absolute money reaching each person**

Factors Affecting Life Expectancy

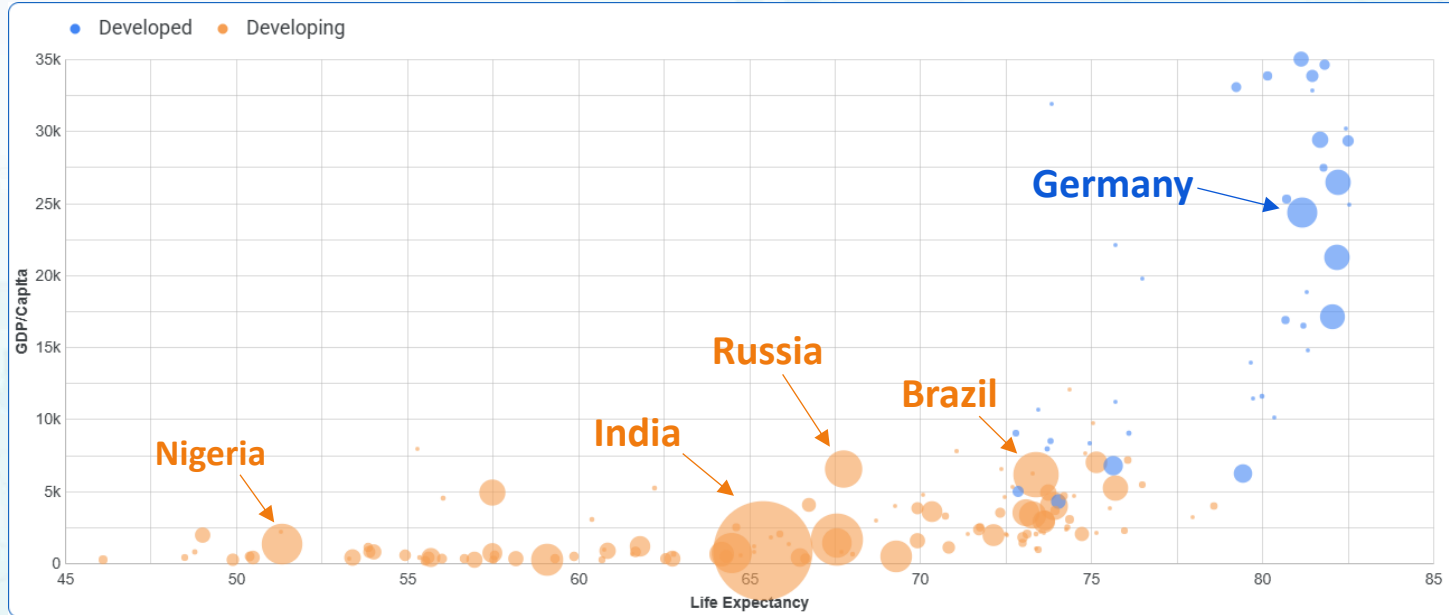
GDP/Capita vs Life Expectancy; Population = Bubble Size



Population size virtually has no correlation with life expectancy

Factors Affecting Life Expectancy

GDP/Capita vs Life Expectancy; Population = Bubble Size



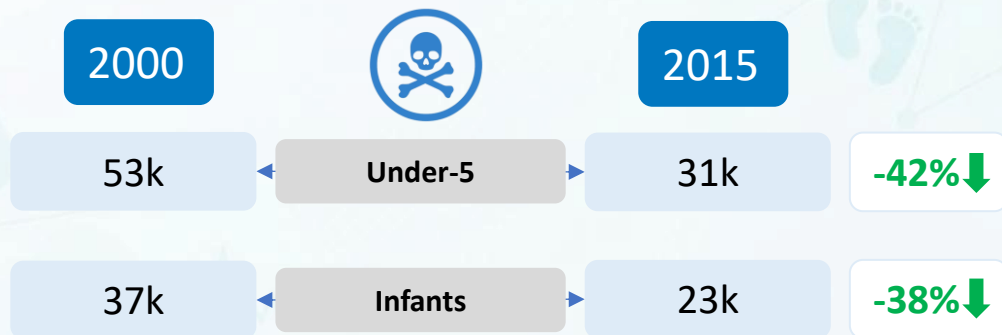
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The Children's Story

Through the UNICEF lens: How did child health improve?

The Youngest Victims



**Averaged across 193 countries*

Key patterns

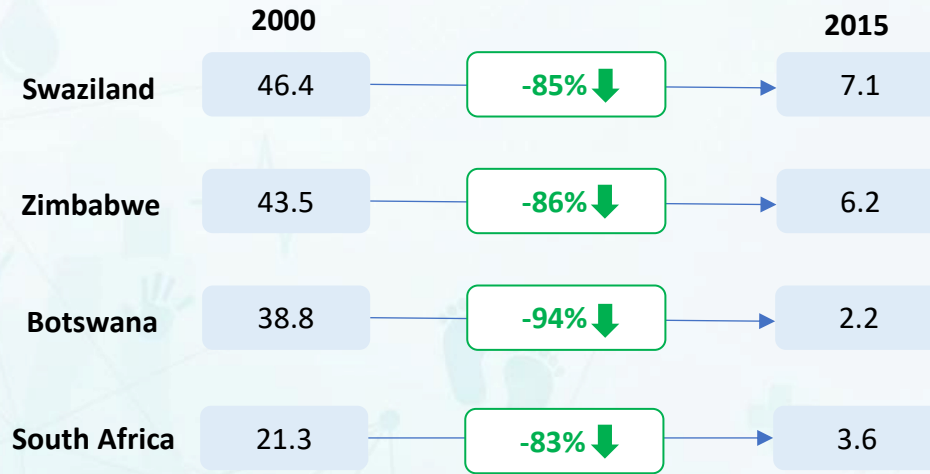
- Mortality measures declined globally since 2000 owing to immunization and healthcare.
- Infant and under-5 mortality remain concentrated in parts of Asia and Sub-Saharan Africa.
- India – still the highest in Infant deaths. ~930k

Conflict and instability disrupt health outcomes

Some countries like Republic of Congo, Chad and Somalia show limited mortality improvement, driven by conflict and political instability

The Battle is not Over

Huge progress has been made with HIV/AIDS deaths but cases still prevail



*Deaths per 1000 births

Top 9 Countries with HIV/AIDS Deaths per 1000 births

	Country	HIV/AIDS
1.	Lesotho	9.3
2.	Swaziland	7.1
3.	Zimbabwe	6.2
4.	Malawi	4.8
5.	Central African Republic	4.4
6.	Equatorial Guinea	4.2
7.	Zambia	4.1
8.	Mozambique	3.9
9.	Nigeria	3.7

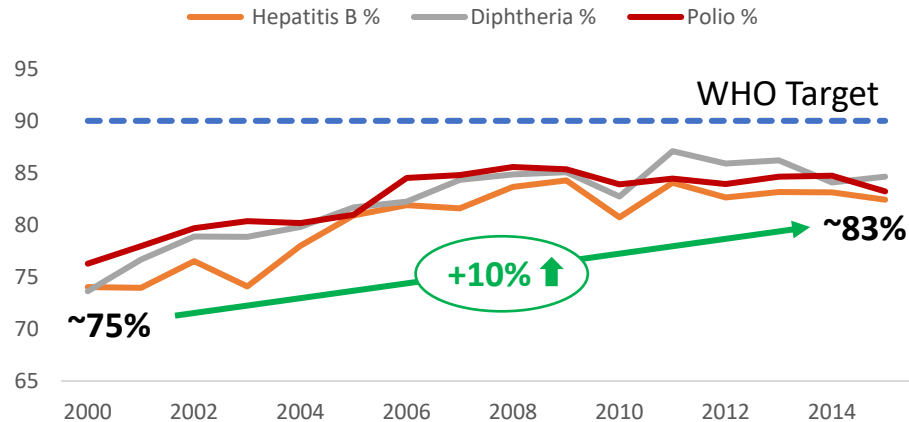
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80% of HIV/AIDS deaths still come from African nations

Immunization Progress

The good news – Immunization coverage has improved thanks to sustained efforts

Immunization coverage (%) vs Year



Key takeaway:

- Coverage increased linearly since 2000 (~10pts).
- Global averages remain below WHO's 90% target (herd immunity threshold).
- Progress slowed after 2008: financial crisis + last-mile constraints.

Many countries have achieved high coverage. Remaining gaps are concentrated in a few hotspots



Unfinished Business

Where should the new Director focus resources for maximum impact?

Recommendations for the Director

Because progress was uneven, and the biggest levers were **education and child health**, we recommend four priorities to **close today's gaps and protect future gains**.

01

Double Down on Sub-Saharan Africa

Evidence: 15/15 bottom countries in Africa across data points

Invest In: HIV/AIDS + maternal health + immunization → **reduce under-5 & adult mortality**

02

Invest in Education & Advisory Policy

Evidence: Schooling shows strongest correlation

Invest In: Girls' education + policy support → **sustained gains**

03

Close Immunization Gaps

Evidence: Global avg still < WHO 90% target

Invest In: local delivery support for Hep B / Diphtheria / Polio hotspots → **prevent outbreaks**

04

Build roadmap towards Next Generational Problems

Evidence: High PM2.5 is linked to poorer health outcomes. Correlation has increased in last 5 years

Invest In: clean-air roadmap (monitoring + clean energy partnerships) → **lower chronic disease burden**



Sources/Annotations

- [UNDP Data for HDI](#)
- [World Bank PM2.5 Air pollution data](#)
- [WHO 90% target](#)
- HIV/AIDS insights
 - [NCBI](#)
 - [CDC](#)
 - [International Aids Journal](#)